Institutional Retention of US MD—Granting Medical School Graduates for Graduate Medical Education: A National, Cross-Specialty Study

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OBJECTIVE: To assess the prevalence and characteristics of institutional retention (IR), i.e., graduates entering a graduate medical education (GME) program at an institution affiliated with their medical school (a "home" residency program), among 5 recent national cohorts of US MD—granting medical school graduates.

DESIGN: This is a retrospective, cross-sectional study that data from the Association of American Medical Colleges Student Records System, Electronic Residency Application Service (ERAS®), and National GME Census in GME Track® databases. Graduate characteristics included GME academic year and specialty. Multilevel logistic regression estimated the odds of IR, reporting adjusted odds ratios (AORs), and 95% confidence intervals (CIs).

SETTING: US MD—granting medical school graduates in academic years 2018 to 2019 through 2022 to 2023 who entered categorical positions in GME programs immediately after graduation, and for whom it was possible to be retained at their undergraduate medical education institution due to an affiliation with a program in their specialty.

PARTICIPANTS: A total of 69,860 (69.8% of 100,155) medical school graduates.

RESULTS: IR prevalence was 24.2% (16,927/69,860), increasing in recent cohorts and varying by specialty (18.8% in emergency medicine to 48.1% in integrated thoracic surgery). Recent graduates (AOR = 1.03, 95% CI = 1.02-1.04), fewer ERAS applications (each vs. >25 ERAS applications; 0 or 1: AOR = 25.48, 95% CI = 20.27-32.03; 2-10: AOR = 6.91, 95% CI = 5.91-8.07; 11-25: AOR = 1.84, 95% CI = 1.74-1.94), and graduates from public (AOR = 1.49, 95% CI = 1.26-1.75) or research-intensive schools (AOR = 1.57, 95% CI = 1.23-2.02) had higher odds of IR. MD—PhD (AOR = 0.65, 95% CI = 0.57-0.73) and other advanced combined MD degree graduates (AOR = 0.86, 95% CI = 0.78-0.95) had lower odds.

CONCLUSIONS: IR has become more common and varies by specialty. The findings highlight some of the many contextual factors shaping undergraduate medical education-to-GME transitions, provide national benchmarking data for US medical schools generally and for GME educators and leaders on a specialty-specific basis. (J Surg Ed 83:104174. © 2026 The Author(s). Published by Elsevier Inc. on behalf of Association of Program Directors in Surgery. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/))

KEY WORDS: undergraduate medical education, graduate medical education, transition to residency, institutional retention

ABBREVIATIONS: UME, undergraduate medical education; GME, graduate medical education; IR, institutional retention; ERAS®, Electronic Residency Application System; AAMC, Association of American Medical Colleges; PGY, postgraduate year

INTRODUCTION

The transition from undergraduate medical education (UME) to graduate medical education (GME) is a complex process for medical schools as senders of graduating students and for residency programs as receivers of graduates, all in the context of individual student-level preferences regarding specialty and location, among

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other considerations. Within this system, institutional retention (IR)—when a medical school graduate enters a residency program affiliated with their graduating institution (a "home" residency program)—reflects factors operating at both the individual and institutional levels. For example, graduates may prefer to remain at their home institution due to existing mentorship and interprofessional networks,[1,2] ongoing research opportunities,[1,3] or local family connections. At the institutional level, IR may occur due to long-standing UME–GME partnerships, alignment of educational missions, and workforce incentives, as institutions may aim to recruit physicians who are more likely to practice in the geographic regions where they complete residency training.[3-7] Thus, GME programs may seek to retain graduates of medical schools with which they are affiliated.

Research to date on the prevalence of IR has largely been limited to single-specialty studies (largely in various surgical specialties that have historically been competitive [i.e., most positions in the specialty offered in a match are filled via the matching process, and most filled positions are secured by US MD senior students who matched to the positions] for entry) with a focus on whether the presence of home residency program in a given specialty is associated with an applicant's likelihood of securing a GME position in the specialty generally, in the geographic region of the applicant' medical school, and/or at the applicant's home institution specifically.[2,8-10] To our knowledge, the prevalence of IR among those US MD–granting school graduates for whom IR was a possibility (i.e., there was a program affiliated with the graduate's medical school in the GME specialty the graduate entered), has never been described across a range of both surgical and nonsurgical specialties. To address this gap, we conducted a retrospective observational study to describe the national prevalence of IR across specialties for 5 recent cohorts of entering residents who were graduates of US MD–granting medical schools.

METHODS

For our retrospective, observational, cross-sectional study, we created a deidentified dataset of our study-eligible population, which included all US MD–granting medical school graduates who, based on National GME Census records,[11] entered Accreditation Council for Graduate Medical Education–accredited GME programs in calendar years 2019 through 2023. Given the complexity of the residency selection process, and the variation in many aspects of both UME and GME programs in the US, a very broad range of individual-level, institutional, and specialty-specific variables are undoubtedly contributory to IR. Based on a review of the literature and other published data on IR,[3,5,6,12-19] we selected a discrete set of variables for this initial exploratory study, utilizing individual-level data from the Association of American Medical Colleges (AAMC) GME Track®,[11] Student Records System,[20] and Electronic Residency Application System (ERAS®) databases[12]; and program-level data from the AAMC GME Track,[11] and the AAMC's Organizational Characteristics Database.[21]

AAMC GME Track®

From GME Track, the primary outcome, IR, was defined as a graduate's entry, in a categorical position, into a residency program affiliated with their medical school. We restricted our study to graduates who entered categorical positions (i.e., positions for all years of required training in a given specialty, from the first postgraduate year until completion of the requisite period of training for specialty board certification eligibility). Residency program–medical school affiliations were based on the graduate's first residency placement and derived from the 2023 GME Track® Program Survey, program names, and sponsoring institution data. We created a dichotomous IR variable for each graduate ("retained" [if their GME program included their medical school among its reported affiliations or if the medical school of graduation was the sponsor of the GME program] vs. "not retained"). Also using GME Track® Program Survey data, we created an 18-category specialty variable (including 9 surgical and 9 nonsurgical specialties) for those specialties in which most or all of the entry-level positions were categorical (rather than advanced) positions and, to ensure adequate sample sizes, for which there were at least 150 graduates in the database who had entered categorical positions in the specialty.

From Student Records System (SRS), we included academic year of graduation and degree type (MD [including BS/BA-MD and accelerated-MD], MD-PhD, or MD in combination with any other non-PhD advanced degree, e.g., MD-MPH, MD-JD). We included graduation year because specialty-specific home program "match rates" were reportedly higher during and shortly after the onset of the COVID-19 pandemic[5,15] (a period encompassed in the graduation year cohorts in our study); we included degree program of graduation because postgraduation outcomes (e.g., specialty for GME) for US MD–granting medical school graduates reportedly differ on the basis of degrees held by the graduate.[14,16]

From ERAS®, we included the number of ERAS® applications as a 4-category variable (0-1, 2-10, 11-25, and >25). Application counts were derived from ERAS records. The ERAS application number categories were

created based on the distribution of ERAS application numbers for graduates in each specialty (these numbers vary widely based on specialty[22]) while also maintaining adequate sample sizes across specialties.

Given the wide range of medical school missions[13] (e.g., prepare physicians to fulfill the needs of the community) that could contribute to our outcome of interest (IR), and work of others on institutional factors associated with GME program geographic proximity to a graduates' medical school,[3] we included 3 school-level characteristics derived from information in the AAMC Organizational Characteristics Database[21]: ownership (public vs. private); community-based designation; and research-intensive designation (defined as those medical schools that appeared in the top 20 schools for federal research expenditure across all 5 years [2019-2023] in which individuals in our study cohort had graduated from medical school).

Data Analysis

Chi-square tests examined bivariate associations and multilevel logistic regression models with graduates nested within medical schools were used to identify variables independently associated with our outcome (IR); 2-sided $p < 0.05$ were considered to be significant. StataNow/SE, 19.5 (StataCorp, College Station, Texas) was used for all analyses.

RESULTS

Our study sample derivation is shown in Figure 1. As shown, of all 100,155 study-eligible graduates, our final study sample included 69,860 graduates (69.8% of 100,155) who (1) entered a categorical position in a nonmilitary GME program (we excluded military GME programs due to limited data about these programs) in the same year as graduation in one of 18 specialties, (2) attended a nonmilitary medical school that had graduating cohorts in all 5 years (ensuring consistent longitudinal representation across medical schools), and (3) had the opportunity to enter a program at their home institution (i.e., there was a home-institution GME program in the graduate's program specialty). Of these 69,860 graduates, 16,927 (24.2%) were retained at their home institutions for GME. This IR proportion varied across the 141 medical schools attended by graduates in the study sample, with each proportion calculated as the average annual percentage (mean, 23.9%; standard deviation, 9.6).

Table 1 presents the characteristics of graduates classified by IR status. IR showed significant variation across each study variable (all, $p < 0.001$). Table 2 describes ERAS application volumes by specialty. More than 90% of graduates who entered the specialties of anesthesiology, emergency medicine, obstetrics and gynecology, general surgery, and other surgical fields (including integrated plastic surgery, vascular surgery, thoracic surgery, neurological surgery, orthopedic surgery, otolaryngology, and urology, were grouped due to small sample sizes)—submitted 25 or more applications. In contrast, graduates who entered the specialties of family medicine, internal medicine (IM), IM/pediatrics, neurology, pathology, pediatrics, and psychiatry submitted a wider range of application numbers. Notably, as also shown in Table 2, there were small numbers of graduates in GME in every specialty category who had submitted 0 or only 1 ERAS® application. The 703 graduates who had submitted 0 or only 1 ERAS® application (as shown in Table 1) included 217 who had submitted 0 and 486 who had submitted 1 ERAS® application.

Expanding on Table 2, Table 3 shows IR status by ERAS® application numbers within specialties. Findings show that across all specialties, at least 49% of graduates who submitted 10 or fewer applications were retained at their home institution (data not shown for IM/pediatrics, neurology, and obstetrics and gynecology due to small N sizes for application categories "0 to 1" and "2 to 10"). IR rates for graduates who submitted 10 or fewer applications were especially high for family medicine (0 or 1 applications, 209/224 [93.3%]; 2-10 applications, 163/223 [66.7%]), IM (0 or 1, 134/149 [89.9%]; 2-10, 164/248 [66.1%]), and pediatrics (0 or 1, 86/93 [92.5%]; 2-10, 68/110 [61.8%]).

Table 4 shows the results of a multilevel logistic regression model estimating the likelihood of IR. As shown, IR retention odds were *higher* for graduates in more recent first postgraduate years (adjusted odds ratio [AOR] = 1.03; 95% confidence interval [CI] = 1.02-1.04); for graduates in the specialties of (each vs. IM) Anesthesiology (AOR = 1.35; 95% CI = 1.25-1.47), Integrated plastic surgery (AOR = 1.48; 95% CI = 1.22-1.81), Integrated thoracic surgery (AOR = 3.34; 95% CI = 2.11-5.27), Integrated vascular surgery (AOR = 1.94; 95% CI = 1.36-2.78), IM/Pediatrics (AOR = 1.47; 95% CI = 1.29-1.68), Neurological surgery (AOR = 1.50; 95% CI = 1.27-1.76), Neurology (AOR = 1.48; 95% CI = 1.29-1.69), Orthopedic surgery (AOR = 1.45; 95% CI = 1.32-1.58), Otolaryngology (AOR = 1.44; 95% CI = 1.26-1.64), Psychiatry (AOR = 1.48; 95% CI = 1.38-1.59), Surgery (AOR = 1.08; 95% CI = 1.04-1.36), and Urology (AOR = 1.19; 95% CI = 1.04-1.36); for graduates who submitted (each vs. >25 ERAS applications), 0 or 1 (AOR = 25.48; 95% CI = 20.27-32.03), 2 to 10 (AOR = 6.91; 95% CI = 5.91-8.07), or 11 to 25 (AOR = 1.84; 95% CI = 1.74-1.94) ERAS applications; for graduates of public (vs. private) medical schools (AOR = 1.49; 95% CI = 1.26-1.75); and for graduates of

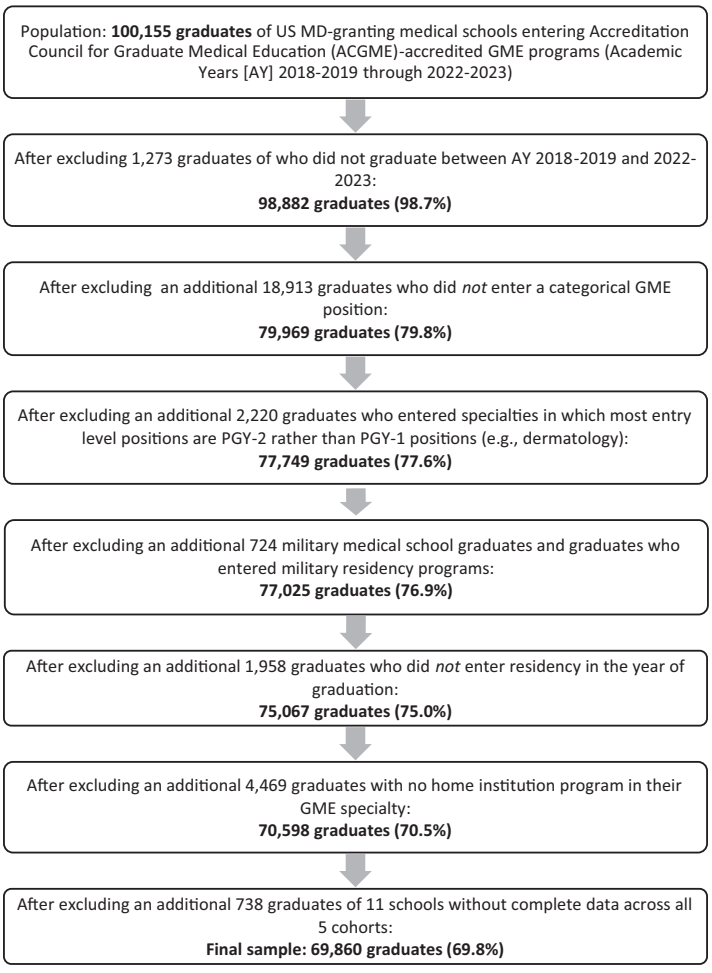


FIGURE 1. Flow diagram for cohort selections and exclusions.

research-intensive (vs. all other) medical schools (AOR = 1.57; 95% CI = 1.23-2.02).

IR retention odds were *lower* for graduates in the specialty categories of (each vs. IM) Emergency medicine (AOR = 0.86; 95% CI = 0.80-0.92), Pathology (AOR = 0.83; 95% CI = 0.70-0.99), and Pediatrics (AOR = 0.69; 95% CI = 0.65-0.74); and for (each vs. MD-degree graduates) MD-PhD (AOR = 0.65; 95% CI = 0.57-0.73) and MD-other advanced (AOR = 0.86; 95% CI = 0.78-0.95) combined degree graduates. IR retention odds were *similar* (each $p > 0.05$) for graduates in the specialties of (each vs. IM) Family medicine and Obstetrics and gynecology; and for graduates of community-based (vs. all other) medical schools.

DISCUSSION

We observed an overall prevalence of IR of 24.2% among 5 cohorts of US MD—granting medical school graduates. The IR prevalence we observed is somewhat higher than that of 15.8% (4126/26,102) reported in a multispecialty study that relied on data in the public domain (e.g., "match" lists published by individual medical schools; such lists generally describe postgraduation plans for an entire graduating class, rather than listing only those graduates who actually secured their positions by matching) for a subset of US Liaison Committee on Medical Education-accredited medical schools' graduates in a range of specialties in 2018 to 2020.[3] Numerous differences in study design and selection criteria for study inclusion are likely contributory to the lower IR prevalence that was observed by Hasnie et al.[3]

In placing our study findings in the context of the existing literature, it is important to note that, although most graduates in our study (all of whom entered GME in categorical positions) had likely secured the categorical positions in the specialties they entered via a matching process (particularly for the surgery specialties we examined that filled the large majority of their categorical positions via a matching process over the years included in our study[23,24]), this was not necessarily

TABLE 1. Study Sample Grouped by Institutional Retention (*N* = 69,860)

	Total Graduates (*N* = 69,860)		Not Retained (*N* = 52,933)		Retained (*N* = 16,927)		p-Value
	N	**%***	***N***	**%†**	***N***	**%†**	
GME PGY-1 year							<0.001
2019	13,876	19.9	10,682	77.0	3194	23.0	
2020	14,032	20.1	10,740	76.5	3292	23.5	
2021	14,561	20.8	10,872	74.7	3689	25.3	
2022	13,963	20.0	10,584	75.8	3379	24.2	
2023	13,428	19.2	10,055	74.9	3373	25.1	
ERAS application no.							<0.001
0-1	703	1.0	91	12.9	612	87.1	
2-10	812	1.2	277	34.1	535	65.9	
11-25	10,136	14.5	6882	67.9	3254	32.1	
>25	58,209	83.3	45,683	78.5	12,526	21.5	
Degree program at graduation‡							0.041
MD	65,445	93.7	49,520	75.7	15,925	24.3	
MD-PhD	1798	2.6	1382	76.9	416	23.1	
Combined MD-other advanced	2617	3.7	2031	77.6	586	22.4	
Specialty (number of schools with affiliated GME program[s])							<0.001
Anesthesiology (108)	3939	5.6	2888	73.3	1051	26.7	
Emergency medicine (131)	7350	10.5	5966	81.2	1384	18.8	
Family medicine (132)	7570	10.8	5408	71.4	2162	28.6	
Integrated plastic surgery (77)	525	0.8	373	71.0	152	29.0	
Integrated thoracic surgery (31)	77	0.1	40	51.9	37	48.1	
Integrated vascular surgery (60)	141	0.2	91	64.5	50	35.5	
Internal medicine (141)	17,186	24.6	13,148	76.5	4038	23.5	
Internal medicine/pediatrics (74)	1164	1.7	769	66.1	395	33.9	
Neurological surgery (106)	825	1.2	588	71.3	237	28.7	
Neurology (100)	1143	1.6	797	69.7	346	30.3	
Obstetrics and gynecology (137)	5170	7.4	4094	79.2	1076	20.8	
Orthopedic surgery (124)	3119	4.5	2268	72.7	851	27.3	
Otolaryngology (101)	1307	1.9	937	71.7	370	28.3	
Pathology (100)	773	1.1	556	71.9	217	28.1	
Pediatrics (136)	7970	11.4	6407	80.4	1563	19.6	
Psychiatry (137)	5407	7.7	3814	70.5	1593	29.5	
Surgery (137)	4907	7.0	3817	77.8	1090	22.2	
Urology (113)	1287	1.8	972	75.5	315	24.5	
School-level							
Research-intensive							<0.001
No	61,969	88.7	47,567	76.8	14,402	23.2	
Yes	7891	11.3	5366	68.0	2525	32.0	
Community-based							<0.001
No	61,280	87.7	46,267	75.5	15,013	24.5	
Yes	8580	12.3	6666	77.7	1914	22.3	
Public ownership							<0.001
No	26,232	37.5	20,637	78.7	5595	21.3	
Yes	43,628	62.5	32,296	74.0	11,332	26.0	

Abbreviation: PGY, postgraduate year.

*Percentages shown are percentages of column total for each variable.

†Percentages shown are percentages of row total.

‡Included in the "MD" group is a small number of accelerated program graduates (*N* = 334). These graduates had an 81.4% (272/334) overall rate of institutional retention.

TABLE 2. ERAS Applications by Specialty ($N = 69,860$)*

Specialty	Total *N* (%)†	0-1: *n* (%)‡	2-10: *n* (%)‡	11-25: *n* (%)‡	>25: *n* (%)‡
Anesthesiology	3939 (5.6)	42 (1.1)	22 (0.6)	168 (4.3)	3707 (94.1)
Emergency medicine	7350 (10.5)	39 (0.5)	30 (0.4)	317 (4.3)	6964 (94.7)
Family medicine	7570 (10.9)	224 (3.0)	223 (2.9)	2635 (34.8)	4488 (59.3)
Internal medicine	17,186 (24.6)	149 (0.9)	248 (1.4)	2792 (16.2)	13,997 (81.4)
Internal medicine/pediatrics	1164 (1.7)	13 (1.1)	6 (0.5)	301 (25.9)	844 (72.5)
Neurology	1143 (1.6)	13 (1.1)	5 (0.4)	127 (11.1)	998 (87.3)
Obstetrics and gynecology	5170 (7.4)	10 (0.2)	7 (0.1)	80 (1.5)	5073 (98.1)
Other surgical specialties§	7281 (10.4)	43 (0.6)	12 (0.2)	58 (0.8)	7168 (98.4)
Pathology	773 (1.1)	14 (1.8)	84 (10.9)	340 (44.0)	335 (43.3)
Pediatrics	7970 (11.4)	93 (1.2)	110 (1.4)	2698 (33.9)	5069 (63.6)
Psychiatry	5407 (7.8)	38 (0.7)	55 (1.0)	535 (9.9)	4779 (88.4)
Surgery	4907 (7.0)	25 (0.5)	10 (0.2)	85 (1.7)	4787 (97.6)

*Chi-square, $p < 0.001$.
†Percentages shown are percentages of column total.
‡Percentages shown are percentages of row total.
§Because of small sample sizes (less than 5) across certain categories of ERAS applications, the following 7 specialties were grouped under "other surgical specialties": integrated plastic surgery, integrated vascular surgery, integrated thoracic surgery, neurological surgery, orthopedic surgery, otolaryngology, and urology.

invariably so. Small numbers of graduates in our study sample may have secured their categorical positions via the National Resident Matching Program Supplemental Offer and Acceptance Program[25] or through other agreements before or after a matching process (for which they may not have had to submit any ERAS applications), including even the most highly competitive specialties. Without actual matching process outcomes data, assumptions cannot be made that any given resident matched to their GME position, or an entire group of residents in a given specialty uniformly secured their GME positions via matching. While GME placement outcomes most closely approximate actual matching process outcomes in competitive (compared to noncompetitive) specialties, even the most highly competitive specialties do not consistently fill all their available positions every year via a matching process.[14,24] This caveat applies to numerous studies in the literature on IR that examined public domain data for residents in various specialties and the medical schools they attended and described such residents as "matched" in the specialty.[3,5,6,26-30] The residents in these studies would be more accurately described as having secured GME positions in the specialty largely through matching, but not as uniformly successfully matched to those positions in the specialty.

The increase in IR in more recent years that we observed is consistent with research showing higher home program "match rates" (i.e., GME placement) during and after COVID-19.[5,15] However, causality cannot be inferred from our retrospective, observational cross-sectional study. During our study period, program signaling[31,32] and geographic preference indicators[33,34] were introduced for ERAS® users. We speculate that these ERAS® modifications may have been among the factors that contributed to an increasing likelihood of IR in more recent years, possibly by more clearly indicating for program directors the extent to which home-program applicants prioritize remaining in the local region, and at their home institutions, for GME.

Our cross-specialty ERAS® application volume data indicates that, although most graduates in our study had submitted >25 ERAS® applications, some graduates who had entered GME in every specialty category we examined had submitted very low numbers of ERAS® applications. Graduates in our study sample (who had all successfully secured GME positions for entry immediately after graduation) who had submitted fewer ERAS® applications were much more likely to be retained at their home institutions, in a model that also controlled for GME specialty. We speculate that those graduates in our study sample who submitted low numbers of ERAS® applications may have had particular levels of confidence in local opportunities, possibly through participation in directed pathways, such as Accelerated MD programs[17,35] which are being offered at an increasing number of medical schools,[36] or through other informal means not captured in currently available data.

Single-specialty reports, utilizing public domain data, have described the prevalence of home program "matching" in various competitive specialties (i.e., specialties that fill the vast majority of positions that are offered in a match, with US senior MD students matching to most of the filled positions) including the surgical specialties of plastic surgery,[28] otolaryngology,[6] and orthopedic surgery,[27] among other specialties. Our study

TABLE 3. Institutional Retention Status by Number of ERAS Applications Submitted Within Specialties (*N* = 69,860)

Number of ERAS Applications Submitted Within Specialty	Total Graduates		Not Retained		Retained		p-Value
	N	%*	*N*	%†	*N*	%†	<**0.001**
Anesthesiology							
0-10‡	64	1.6	17	26.6	47	73.4	
11-25	168	4.3	95	56.5	73	43.5	
>25	3707	94.1	2776	74.9	931	25.1	
Emergency medicine							<0.001
0 or 1	39	0.5	6	15.4	33	84.6	
2-10	30	0.4	10	33.3	20	66.7	
11-25	317	4.3	204	64.4	113	35.6	
>25	6964	94.7	5746	82.5	1218	17.5	
Family medicine							<0.001
0 or 1	224	3.0	15	6.7	209	93.3	
2-10	223	2.9	60	26.9	163	73.1	
11-25	2635	34.8	1762	66.9	873	33.1	
>25	4488	59.3	3571	79.6	917	20.4	
Internal medicine							<0.001
0 or 1	149	0.9	15	10.1	134	89.9	
2-10	248	1.4	84	33.9	164	66.1	
11-25	2792	16.2	1841	65.9	951	34.1	
>25	13,997	81.4	11,208	80.1	2789	19.9	
Internal medicine/pediatrics							<0.001
0-25§	320	27.5	158	49.4	162	50.6	
>25	844	72.5	611	72.4	233	27.6	
Neurology							<0.001
0-25§	145	12.7	67	46.2	78	53.8	
>25	998	87.3	730	73.1	268	26.9	
Obstetrics and gynecology							<0.001
0-25§	97	1.9	58	59.8	39	40.2	
>25	5073	98.1	4036	79.6	1037	20.4	
Other surgical specialties‖							<0.001
0 or 1	43	0.6	7	16.3	36	83.7	
2-10	12	0.2	7	58.3	5	41.7	
11-25	58	0.8	29	50.0	29	50.0	
>25	7168	98.4	5226	72.9	1942	27.1	
Pathology							<0.001
0-10‡	98	12.7	44	44.9	54	55.1	
11-25	340	44.0	256	75.3	84	24.7	
>25	335	43.3	256	76.4	79	23.6	
Pediatrics							<0.001
0 or 1	93	1.2	7	7.5	86	92.5	
2-10	110	1.4	42	38.2	68	61.8	
11-25	2698	33.9	2041	75.6	657	24.4	
>25	5069	63.6	4317	85.2	752	14.8	
Psychiatry							<0.001
0 or 1	38	0.7	7	18.4	31	81.6	
2-10	55	1.0	21	38.2	34	61.8	
11-25	535	9.9	333	62.2	202	37.8	
>25	4779	88.4	3453	72.3	1326	27.7	
Surgery							<0.001
0-10‡	35	0.7	18	51.4	17	48.6	
11-25	85	1.7	46	54.1	39	45.9	
>25	4787	97.6	3753	78.4	1034	21.6	

*Percentages shown are percentages of column total for each variable.
†Percentages shown are percentages of row total.
‡Categories "0 or 1" and "2 to 10" were combined due to some cells having N sizes of 5 or less.
§Categories "0 or 1," "2 to 10," and "11 to 25" were combined due to some cells having N sizes below 5.
‖"Other surgical specialties" included integrated plastic surgery, integrated vascular surgery, integrated thoracic surgery, neurological surgery, orthopedic surgery, otolaryngology, and urology.

TABLE 4. Multilevel Logistic Regression Results Predicting the Likelihood of Institutional Retention ($N = 69,860$)

	Adjusted Odds Ratio	95% Confidence Interval	p-Value
GME PGY-1 year, per more recent year	1.03	1.02-1.04	<0.001
Number of ERAS applications			
0 or 1	25.48	20.27-32.03	<0.001
2-10	6.91	5.91-8.07	<0.001
11-25	1.84	1.74-1.94	<0.001
>25	Reference	Reference	
Degree program at graduation			
MD	Reference	Reference	
Combined MD-PhD	0.65	0.57-0.73	<0.001
Combined MD-other advanced	0.86	0.78-0.95	0.003
Specialty			
Anesthesiology	1.35	1.25-1.47	<0.001
Emergency medicine	0.86	0.80-0.92	<0.001
Family medicine	1.01	0.95-1.08	0.686
Integrated plastic surgery	1.48	1.22-1.81	<0.001
Integrated thoracic surgery	3.34	2.11-5.27	<0.001
Integrated vascular surgery	1.94	1.36-2.78	<0.001
Internal medicine	Reference	Reference	
Internal medicine/pediatrics	1.47	1.29-1.68	<0.001
Neurological surgery	1.50	1.27-1.76	<0.001
Neurology	1.48	1.29-1.69	<0.001
Obstetrics and gynecology	1.01	0.94-1.09	0.770
Orthopedic surgery	1.45	1.32-1.58	<0.001
Otolaryngology	1.44	1.26-1.64	<0.001
Pathology	0.83	0.70-0.99	0.037
Pediatrics	0.69	0.65-0.74	<0.001
Psychiatry	1.48	1.38-1.59	<0.001
Surgery	1.08	1.00-1.17	0.049
Urology	1.19	1.04-1.36	0.014
School-level			
Research-intensive			
No	Reference	Reference	
Yes	1.57	1.23-2.02	<0.001
Community-based			
No	Reference	Reference	
Yes	0.95	0.78-1.16	0.621
Public ownership			
No	Reference	Reference	
Yes	1.49	1.26-1.75	<0.001

Abbreviation: PGY, postgraduate year.

provides a different perspective on IR, demonstrating that several surgical specialties are characterized by relatively high IR rates compared to various other nonsurgical specialties examined. Given the small number of GME programs and GME positions in many surgical specialties, along with the relatively higher competitiveness for specialty entry, our findings may not necessarily be surprising. IR in surgical specialties (which fill almost all categorical positions via matching processes) reflects both program director and applicant preferences. Program directors may have had much more extensive opportunities to get to know their "home" medical school applicants over an extended period of time and have a higher degree of confidence in those applicants' commitment to the specialty—a particular consideration in some surgical specialties with relatively high rates of attrition during GME.[37] There are pros and cons for applicants to consider in remaining at their home programs for GME.[38] Among the "pros," applicants may prefer their home programs due to established relationships with leaders in their chosen specialties through clinical and research endeavors during medical school,[1-3] and they can "hit the ground running" at the start of GME in an already familiar setting.[38] Across a wide range of specialties, graduates who remained at their home institutions for GME were less likely to be rated by their program directors as failing to meet expectations in the first 6 months of residency.[39]

Our observations about the lower prevalence of IR, and lower odds of IR, for MD-other advanced degree and MD-PhD graduates (compared to MD graduates) extends the evidence that postgraduation trajectories of such dual-degree holders differ from their MD-degree holder counterparts.[14,16,18,40] While it was beyond the scope of our study to explore potential contributory factors to these observations, we speculate that these dual-advanced degree program graduates may have particular career priorities not necessarily met by their "home" GME programs.

Our observations of higher IR prevalence for graduates of research-intensive medical schools and of public (vs. private) medical schools align with findings of a multispecialty study that "matched" graduates (i.e., graduates placed in GME) of public schools and of research-intensive medical schools, were more likely (in a model that controlled for specialty), to have entered programs in very close geographic proximity to their medical schools (i.e., at their home institution or within the same state).[3] Our medical-school ownership IR observation provides an additional perspective regarding the extent to which public medical schools may retain their graduates for local workforce needs. We speculate that our observation regarding higher IR prevalence, and higher IR odds among graduates of research-intensive medical schools, might reflect, at least in part, particular research-focused career priorities of these schools' graduates that may have been nurtured through close working relationships with faculty during medical school.

Our study has several strengths. To our knowledge, this is the first cross-specialty study of IR at a national level. While there is not, de novo, any "right" proportion (for any given medical school, or for any given GME specialty) of graduates that should be retained at their home institutions for GME, our data may be useful for benchmarking purposes for both medical schools and for GME educators on a specialty-specific basis. Unlike most other studies in the literature on IR that utilized public domain data only to identify graduates in GME positions in the specialty and the medical schools attended by these graduates,[3,29,30,41] our study utilized National GME Census data as reported by GME program directors and other AAMC data as reported by medical school registrars in SRS. Given the size and scope of our national dataset, we were also able to examine IR for a broad range of specialties, including several specialties with relatively small GME workforces (e.g., integrated programs in plastic surgery, vascular surgery, and thoracic surgery) and to identify several other variables, in a model that controlled for specialty, that were independently associated with IR. The prevalence of IR of 24.2% that we observed in our study sample generally aligns with the prevalence of IR reported for aggregate data for all US MD−granting medical school graduates entering all position types on an annual basis over the same 5-year period of 23.3% to 24.7%,[19] suggesting that our observations in our restricted study sample may be broadly applicable to US MD−granting medical school graduates generally.

IR may reflect a broader tension within the UME−GME continuum between continuity within institutions and mobility across them. Retaining home graduates may support educational continuity, strengthen mission alignment, and reduce onboarding burden. However, high retention rates may also unintentionally limit cross-institutional mobility in a nationally distributed GME system.

This initial exploratory study also has numerous limitations. We did not control for many other variables that contributed to our outcome. Unmeasured institutional norms and historical UME−GME relationships, not fully captured in available national datasets, may be contributory to IR. Unmeasured graduate preferences (e.g., family location, dual-career constraints, geographic ties) and many unmeasured aspects of both specialty-specific and program-specific resident selection processes also contributed to our outcome. As we excluded graduates who entered GME specialties that predominantly or exclusively offer advanced positions for entry (e.g., Ophthalmology, Dermatology, Diagnostic radiology, Physical medicine and rehabilitation), our results may not generalize to those specialties. Finally, as our study sample was limited to US MD−granting medical school graduates, our results also may not generalize to DO graduates or to international medical graduates. It is important to note that IR is, by definition, not an option for medical school graduates, who comprised nearly 1 in 4 residents (37,109/158,031; 23.5%) in the Accreditation Council for Graduate Medical Education−accredited GME workforce in 2023 to 2024.[42]

CONCLUSIONS

Among recent cohorts of US MD−granting medical school graduates, we observed an overall prevalence of IR of 24.3% across 5 years. IR prevalence varied by specialty and increased among more recent years' graduates.

ETHICAL APPROVAL

This study was reviewed and approved by the Association for Institutional Research Institutional Review Board. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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